

INCOME & PROBLEM GAMBLING

An Analysis of Economic Status and Gambling Harm Among Youth

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Economics of Crime | Wake Forest University

GIBS Youth Survey

1997 – 1999

n = 534

Dataset

Research Question & Motivation

Does household income predict the severity of problem gambling?

Motivation

Problem gambling imposes significant social costs — especially on vulnerable youth populations who lack financial buffers against gambling losses.

Hypothesis

Lower-income individuals face greater financial stress and risk-seeking behavior, leading to higher PROB12MO scores (0–10 scale).

Literature Review — Overview

10 peer-reviewed articles grouped across three themes:

Youth Risk Factors

Richard & King (2023)

Calado et al. (2017)

Savolainen et al. (2021)

Online Gambling & Access

Canale et al. (2016)

Elton-Marshall et al.
(2016)

Income & Gambling Harm

Welte et al. (2015)

Currie et al. (2006)

Wardle et al. (2019)

Strumpf (2003)

Tozzi et al. (2023)

Richard & King (2023) — Focus Article #1

"Annual Research Review: Emergence of Problem Gambling from Childhood to Emerging Adulthood"

Method

Systematic review synthesizing findings from 45 peer-reviewed articles Focuses on young children up to age 25

Key Findings

Early stage exposure such as loot boxes in video games, online casino games, and fantasy sports. These correlate to elevated problem gambling risk. Impulsiveness and emotional behavior key traits. Males disproportionately affected.

Limitation

Mostly cross-sectional studies — limits causal inference. Findings were captured in a snapshot in time rather than following kids over years.

Relevance to Our Project

Demonstrates that online exposure lowers barriers and normalizes gambling to youth. Directly supports our focus on youth populations.

Canale et al. (2016) — Focus Article #2

"Impact of Internet Gambling on Problem Gambling Among Adolescents in Italy"

5x

higher likelihood for a gambling problem for online compared to offline

Data:

ESPAD Italia 2013: large, nationally representative sample of Italian high school students aged 15–19.

Findings:

Non-traditional family structures and higher perceived financial status both increase gambling severity. Remote platforms further add to the risk.

Limitation:

Self-reported data likely be underreported. Illegal sportsbooks hard to track.

Relevance:

Establishes internet accessibility itself as a primary risk factor. Supports our argument that unregulated digital environments are especially dangerous for youth.

Welte et al. (2015) — Focus Article #3

"Gambling and Problem Gambling in the United States: Changes Between 1999 and 2013"

"Problem gambling rates declined as socio-economic status increased — economic conditions and financial stress play a central role."

Method: Nationally representative U.S. survey data compared across two time points: 1999 and 2013.

Limitation: Cross sectional data prevents causal conclusions. External economic shocks like recessions may have influenced trends.

Findings: Problem gambling was concentrated among lower income individuals and younger populations. Problem accelerated due to easier access to gambling over time

Relevance: Directly supports our hypothesis: income and financial stress are key drivers of gambling severity.

Strumpf (2003) — Focus Article #4

"Illegal Sports Bookmakers"

Method

Analyzed arrest records from six illegal sports bookmakers operating during the 1990s. Applied regression equations to financial and operational data from these underground betting operations.

Limitation

Narrow data scope: conclusions drawn from only six bookmaking operations, limiting generalizability to the broader illegal sports betting market.

Key Findings

Illegal bookmakers engage in price discrimination based on geography and individual betting patterns. Risk-adjusted profits are actually lower than legal sportsbooks — challenging the assumption that underground operators earn greater returns.

Relevance to Our Project

Provides direct economic grounding for unregulated gambling markets. Establishes the structural incentives and constraints facing illegal operators — context for understanding black-market gambling environments.

Tozzi et al. (2023) — Focus Article #5

"Determinants of Problem Sports Betting Among College Students"

Method

Survey of college students analyzed via partial least squares structural equation modeling (PLS-SEM). Grounded in behavioral economic theory to identify psychological and situational drivers of problem betting.

Key Findings

Young adults tend to chase losses and bet beyond their means. Online betting environments amplify these impulsivity-driven behaviors. Accessibility of digital platforms compounds the risk of developing problematic gambling habits in this demographic.

Limitation

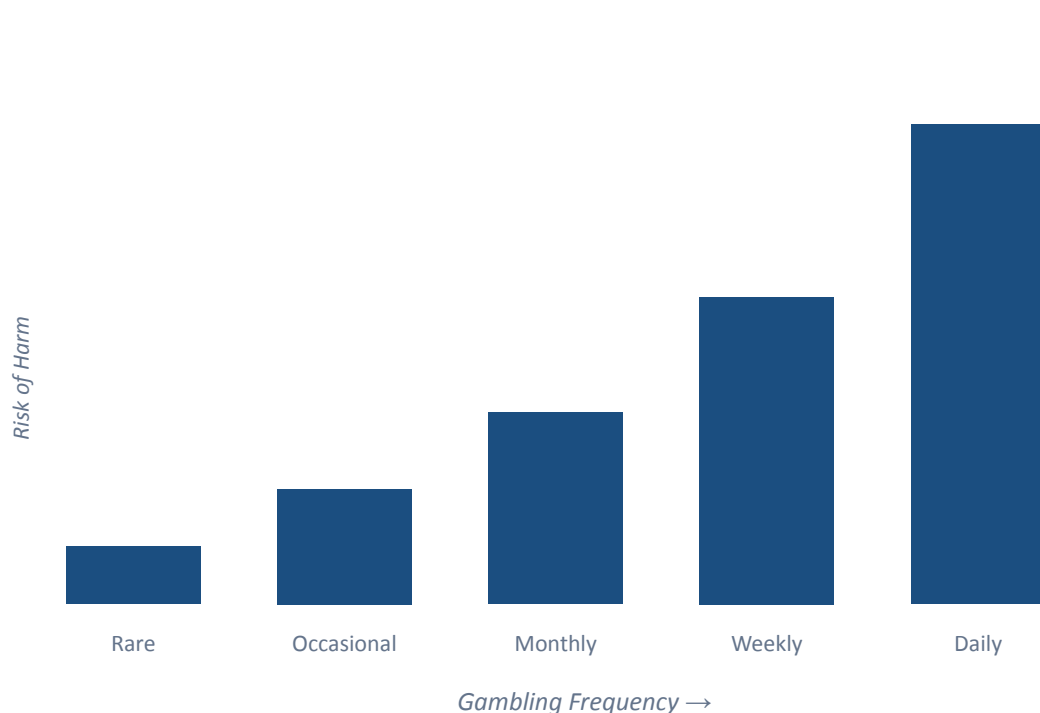
Cross-sectional convenience sample limits generalizability and prevents causal conclusions. Varying legal status of sports betting across U.S. states introduces inconsistency across respondents' environments.

Relevance to Our Project

Directly connects college-aged gambling behavior to economic decision-making. Reinforces our core argument that income, financial stress, and online access are central factors in problem gambling severity.

Currie et al. (2006) — Focus Article #6

"Risk of Harm Among Gamblers as a Function of Level of Participation"



n = 19,000+

Nationally representative Canadian dataset — strong statistical power.

Risk Curve

Harm increases steadily with gambling frequency and expenditure, not just at extreme levels.

Income Share

Gambling expenditure as % of income is a stronger predictor of harm than absolute spending.

Model Implication

Directly motivates use of PROB12MO as a continuous DV and INCOME as the key regressor.

Supporting Literature — Additional Articles

Calado et al. (2017)

Systematic review: 0.2–12.3% of youth meet problem gambling criteria; males, impulsivity, and substance use are primary risk factors.

Savolainen et al. (2021)

Cross-national study (4 countries): conforming to online gambling group norms significantly increases youth problem gambling risk.

Elton-Marshall et al. (2016)

41% of Canadian youth gambled recently despite legal restrictions; online gambling far more prevalent among males.

Wardle et al. (2019)

Gambling harm exists on a continuum — even low-level gamblers face consequences; financially vulnerable populations bear disproportionate harm.

Literature Synthesis — Connecting to Our Project

01

Youth Vulnerability:

Multiple studies confirm adolescents face elevated gambling risk due to impulsivity, peer influence, and online exposure — making the GIBS youth sample ideal.

02

Online Access as Risk:

Canale (2016) and Richard & King (2023) establish digital accessibility as a gateway — relevant to our unregulated market context.

03

Income as a Driver:

Welte (2015) and Wardle (2019) directly show that lower SES predicts greater gambling harm — our core hypothesis.

04

PROB12MO as DV:

Currie (2006)'s continuous risk approach motivates using PROB12MO (0–10) rather than a binary problem-gambling indicator.

Data — Gambling Impact and Behavior Study (GIBS), 1997–1999

Youth Survey | Nationally representative | Ages 12–17 | n = 534

Variable	Type	Description	Mean / Range
PROB12MO	Dependent (continuous)	Gambling problems in past 12 months	0.112 / 0–6
INCOME	Independent (continuous)	Household income level	Varies
J3 (Mental Health)	Control (discrete)	Self-reported mental health status	1.273 / 1–3
A1_ (Gender)	Control (discrete)	Respondent gender (1=M, 2=F)	1.504 / 1–2
Age / Grade (A7_)	Control (discrete)	Highest grade attended	3.318 / 3
Race	Control (discrete)	Racial category	2.022 / –3–3
Region	Control (discrete)	U.S. Census region	2.262 / 1–4

Descriptive Statistics

Variable	N	Mean	Std. Dev.	Min	Max
PROB12MO	534	0.112	0.473	0	6
J3 (Mental Health)	534	1.273	0.516	1	3
A1_ (Gender)	534	1.504	0.500	1	2
A2_R (Race)	534	2.022	0.419	-3	3
Region	534	2.262	1.116	1	4

Low Avg.

Mean PROB12MO = 0.112 — most youth show zero or near-zero gambling problems, consistent with a low-prevalence population.

Right Skew

Max = 6 with SD = 0.473 signals a right-skewed distribution — a small subset reports severe gambling problems.

Gender Split

Mean A1_ $\approx$ 1.50 reflects a near-even male/female split in the sample (1 = male, 2 = female).

Empirical Model

$$\text{PROB12MO} = \beta_0 + \beta_1(\text{INCOME}) + \beta_2(\text{Gender}) + \beta_3(\text{Age}) + \beta_4(\text{Race}) + \beta_5(\text{Region}) + \varepsilon$$

Estimator:

OLS regression — PROB12MO treated as a continuous outcome variable following Currie et al. (2006).

Key Variable:

β_1 (INCOME): Expected to be negative — higher income → fewer gambling problems due to lower financial stress.

Controls:

Gender, age/grade level, race, and region control for demographic heterogeneity in gambling exposure and behavior.

Data Source:

GIBS Youth Survey (1997–1999), accessed via ICPSR/NAHDAP. Cross-sectional design with $n = 534$ youth respondents.

Software:

Stata — tabstat, regress, robust standard errors, VIF diagnostics.

Limitation:

Cross-sectional data prevents causal claims; income and gambling behavior may be jointly determined.

Demographic Regression Analysis

$$\text{Model: PROB12MO} = \beta_0 + \beta_1(\text{Race/Ethnicity}) + \beta_2(\text{Region}) + \beta_3(\text{Gender}) + \varepsilon$$

Focus

Examines specific demographic control variables — Race/Ethnicity, Region, and Gender — and their individual impact on gambling problems over the past 12 months.

Key Finding

Race/Ethnicity and Gender are significant predictors of gambling problem severity. Region was not statistically significant. These demographics shape exposure and vulnerability to gambling harm.

Key Variable	Coefficient	p-value	Result
Race/Ethnicity	0.080	0.002	Significant
Region	0.007	0.714	Not Significant
Gender	-0.107	0.008	Significant

Race/ethnicity and gender each had a statistically significant effect on gambling problem severity — region did not. Demographic factors shape gambling outcomes independent of income.

Expected Results & Conclusion

Based on our literature review and model, we anticipate:

Negative β_1 :

→ A statistically significant negative coefficient on INCOME, indicating lower-income youth experience more gambling problems.

Gender Effect:

→ Males expected to show higher PROB12MO scores, consistent with findings across Calado, Welte, and Elton-Marshall.

Policy Relevance:

→ Results would support income-targeted interventions and stricter age/access controls on online gambling platforms.

Understanding the income–gambling relationship is essential for designing effective, equitable public health responses to youth gambling harm.

Thank You

Questions & Discussion

Dataset: Gambling Impact & Behavior Study (GIBS), 1997–1999 — ICPSR/NAHDAP

DV: PROB12MO (0–10) | **Key IV:** INCOME

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